Efficacy and Safety of Velotrin in Type 2 diabetes: A Phase 3 Randomised, Double-Blind, Placebo-Controlled Trial

Authors: Nakamura A, Rao B, Lindqvist C et al. **Trial registration:** SYN-NCT-3023199

Note: Entirely fictional study with invented drug and data, created for training. Do not use for clinical decisions.

Abstract

Background: Velotrin (VLT) is a once-daily oral SGLT2 inhibitor under investigation for type 2 diabetes. **Methods:** In this multicentre trial, 642 adults were randomised 1:1 to Velotrin or placebo for 52 weeks, stratified by baseline severity and region. The primary endpoint was HbA1c change from baseline at week 26. **Results:** The primary endpoint was -1.18 with Velotrin versus -0.41 with placebo (p <0.001). Genital mycotic infection occurred in 6.2% of Velotrin patients and 1.9% of placebo patients. **Conclusions:** Velotrin met its primary endpoint with a manageable safety profile in type 2 diabetes.

Methods

Eligible participants were aged 18 to 75 years with a confirmed diagnosis of type 2 diabetes and provided written informed consent. Exclusion criteria included pregnancy, severe hepatic impairment (Child-Pugh C), and concurrent use of strong CYP3A4 inducers. Randomisation used an interactive web-response system. The analysis followed the intention-to-treat principle with multiple imputation for missing data.

Results

Characteristic	Velotrin	Placebo
Participants (n)	321	321
Mean age (years)	60.1	61.7
Female (%)	40	45
Primary endpoint	-1.18	-0.41
p-value	<0.001	-
Genital mycotic infection (%)	6.2	1.9
Discontinuation due to AE (%)	3.3	1.1

Serious adverse events were reported in 11.8% of the Velotrin group and 6.3% of the placebo group. There were 5 deaths in the Velotrin arm and 6 in the placebo arm, none judged related to study treatment.

Discussion

The effect size observed with Velotrin is clinically meaningful. Limitations include the 52-week duration, under-representation of patients aged over 75 years, and limited ethnic diversity. Long-term safety beyond one year requires an open-label extension, which is ongoing.

References

1. Synthetic reference list omitted. 2. Illustrative Guidelines on Management, 2025.